TIPA Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Safe LLM-Based Troubleshooting for Incident Response			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication (), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Technology Field 2 is optional*	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Internet SW		
Concept of Proposed Technology	We propose LLM-based troubleshooting agents that can diagnose and mitigate incidents in modern network and cloud systems under realistic operational constraints. The agent operates interactively in a multi-turn manner: it issues read-only diagnostic probes and, when justified, state-changing repair actions; observes system feedback; and iterates until the target service is restored to a healthy state. Off-the-shelf prompting is unreliable for incident response in modern systems because troubleshooting is inherently stateful and high-stakes [1,2,3,4,5]. Each action alters subsequent observability, and plausible but incorrect interventions can silently degrade healthy traffic or exacerbate the outage. To address this challenge, we propose ***fine-tuning LLMs with reinforcement learning with verifiable rewards (RLVR) for troubleshooting***, training the model directly from interactive outcomes and therefore improving its troubleshooting capabilities. This approach teaches the agent to balance evidence gathering with targeted interventions, and to prefer safe recovery strategies under operational constraints. We focus on two technical innovations. First, diagnostics are treated as first-class decisions whose			

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	value is largely downstream. Accordingly, our objective, compatible with Group Relative Policy Optimization (GRPO) [6], propagates recovery and non-regression signals back to earlier diagnostic probes, enabling the model to learn when to probe, what to probe, and when to stop probing. Second, because actions are inherently side-effectful, we explicitly incorporate safety into both the learning signal and runtime guardrails through safe RL [7,8,9,10], ensuring that the agent can restore service reliably without disrupting components that are already functioning correctly. Taken together, this project aims to transform LLM agents into reliable operators for real-world incident response. The resulting agents can reason under uncertainty, gather appropriate evidence before acting, and execute repairs that are both effective and safe. By combining troubleshooting-specific RL fine-tuning with explicit non-regression and safety constraints, we seek to achieve fast and dependable recovery without collateral damage to the system.
Rationale for Joint R&D Project	A joint R&D structure is necessary because the proposed technology lies at the intersection of frontier machine learning methods and real operational system behavior. The NSAI (Networked Systems and AI) Lab at UIUC brings expertise in developing large-scale and practical learning-enhanced systems that operate under real-world constraints. A Korean SME partner can provide realistic operational interfaces, such as network and cloud stacks, that determine whether an agent is practically deployable. In addition, such a partner can supply incident patterns and deployment constraints that are difficult to capture using generic public benchmarks alone.
Keywords	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 1. Large Language Models 2. Incident Response Automation 3. Reinforcement Learning with Verifiable Rewards 4. Safety

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	Our R&D objective is to develop reliable LLM-based troubleshooting agents capable of diagnosing and mitigating incidents in modern network and cloud systems through interactive tool use. The agent is designed to learn how to gather relevant evidence, select safe and targeted interventions, and restore services to a healthy state under realistic observability noise and operational constraints, while producing auditable action traces suitable for real-world operations. To achieve this objective, we will develop troubleshooting-specific RLVR methods that appropriately reward effective diagnostics and safe recovery. We will validate these methods in realistic incident environments and through pilot integrations with the Korean SME's operational stack. The end goal is a deployable prototype that measurably improves incident response outcomes without increasing regressions or violating safety policies.			
R&D Target Specifications	Our R&D target specifications are to achieve high incident recovery success in representative benchmark and pilot environments, while keeping non-regression and safety violations near zero and meeting practical deployability requirements, including low decision latency and fully auditable action logs.			
R&D Plan	We will begin by defining the incident scope, operational constraints, safety and non-regression policies, and the tool interfaces available to the agent. Next, we will develop and train troubleshooting-specific RLVR methods that reward effective diagnostics and safe recovery based on interaction outcomes, and we will integrate runtime guardrails to prevent unsafe actions. Finally, we will integrate the prototype with the Korean SME's monitoring and control surfaces, conduct staged pilot deployments to measure recovery success and regressions, and iterate toward a deployable package accompanied by documentation and technology transfer.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The NSAI Lab at UIUC will lead the core R&D on troubleshooting-specific reinforcement learning fine-tuning, including method design, implementation, and rigorous evaluation to ensure that the agent learns reliable diagnostic and repair behaviors from interactive outcomes. The lab will also define success metrics, develop safety-aware learning and validation protocols, and deliver a working prototype with reproducible training and evaluation pipelines, along with clear technical documentation to support technology transfer.
Expected R&R of Korean SME Partner	The SME will provide realistic operational requirements, incident types, constraints, and access to the relevant system interfaces needed to make the agent practical. The SME will support integration with its tooling and workflows, supply representative scenarios and validation criteria, and run staged pilots to measure impact on recovery success. The SME will also provide iterative feedback to drive the prototype toward production readiness and commercialization.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project will deliver a deployable prototype of an LLM-based troubleshooting agent that can interact with real network and cloud tooling to diagnose incidents, execute safe mitigations, and restore services while producing fully auditable action traces. From a technical perspective, the project will produce validated reinforcement learning fine-tuning methods tailored to incident response. These methods will improve diagnostic decision-making, robustness under noisy observability, and safety and non-regression behavior, and will be accompanied by a reproducible evaluation suite and incident scenarios grounded in operational reality. From an operational perspective, the expected impact is a reduced operator burden during incident response, enabling Korean SMEs to deliver higher reliability and lower downtime costs. More broadly, the project will accelerate the commercialization of safety-first LLM-based troubleshooting capabilities in production environments.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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